LOCAL BUSINESS SUPPORT RECOVERY

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Today's Agenda

Context & Research Site

Why Titan Plaza?

- Located on Av. Boyacá con Calle 80, Bogotá — one of the most visited malls in the city
- 4 floors, 250 stores, daily 8am–10pm
- Attracts middle-income population (strata 3–5)
- Contains a mix of large chains + small independent businesses
- Ideal case study: concentrated, observable, real-world environment

The Problem

Small Businesses Are Struggling Post-Pandemic

- 41.18% of mall visitors spend less than before the pandemic
- Foot traffic is unpredictable: Sunday brings 2x more visitors than Saturday
- Small stores operate as isolated agents - no real-time data
- Large chains (Falabella, Éxito) dominate through digital tools; small shops rely on foot traffic alone
- 80% of small stores report fast/very fast inventory rotation → stockout risk is high

Problem Context:

Restrictions and Shift in Scope

Environmental Boundaries & Shifts:

- The project's core objective was refined to focus explicitly on closing the informational gap for small stores operating in isolation.
- The systemic issue crystallized into severe operational symptoms: low buyer conversion rates despite high foot traffic, and frequent stockouts.

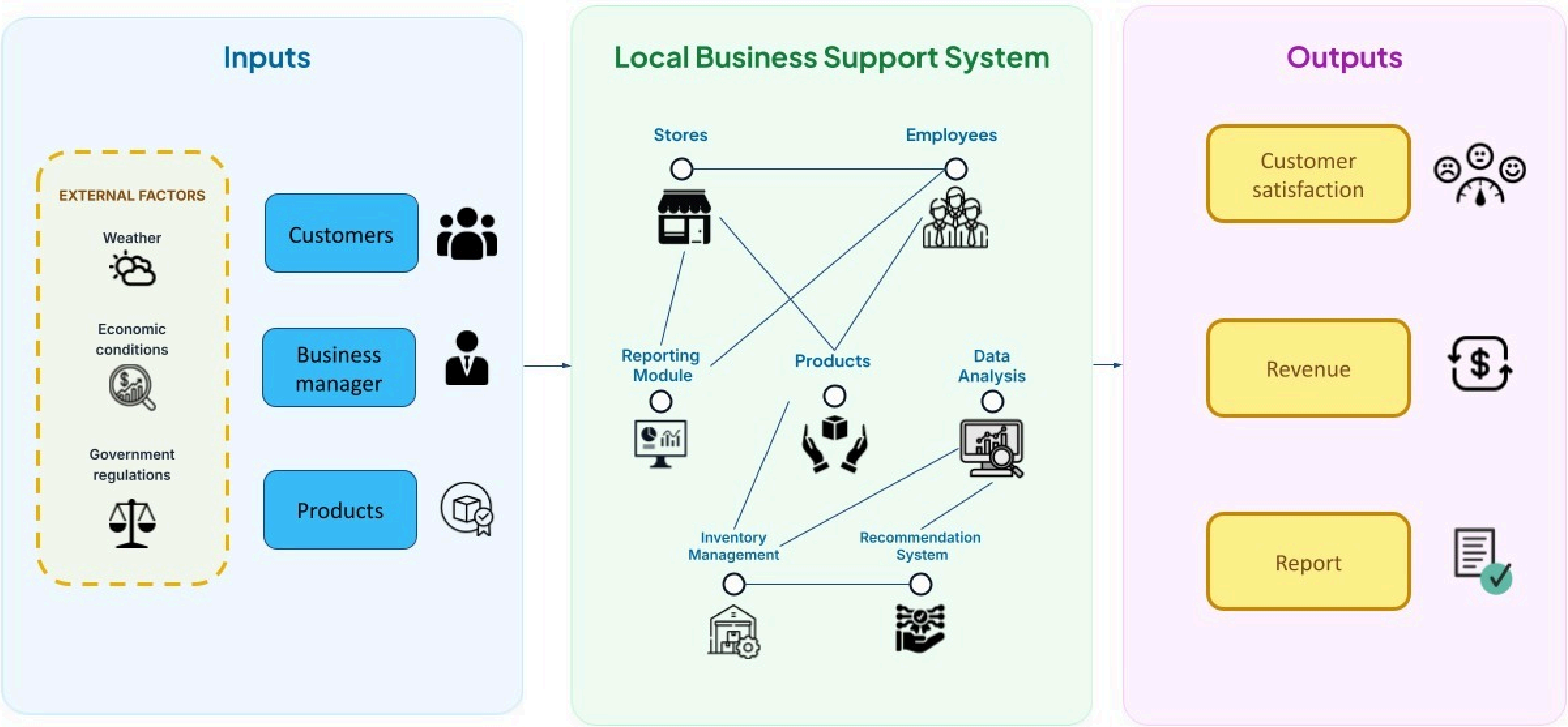
Symptoms & Disequilibrium:

- Physical store capacity and inventory levels reach critical thresholds during weekend peak periods.
- Severe informational entropy emerging due to manual, unstructured inventory management

Project Evolution and Integration

Workshop 1	Systems Analysis	Field-based empirical data collection at Titan Plaza, quantitative surveys with customers and employees, and identification of the initial structural disequilibrium
Workshop 2	System Design	Application of abstraction and modular modeling to represent business interactions, inventory flows, and customer behavior
Workshop 3	Robust Design and Management	Project boundary refinement based on realistic environmental and connectivity constraints
Workshop 4	Dynamic Simulation	Computational and conceptual validation of systemic stability, sales behavior, and inventory resilience

System Diagram



Simulation Architecture

How We Modeled the System

Behavioral Simulation	Process-Oriented Simulation
Adaptive interactions	Sequential operational workflow
Emergent behaviors	Customer arrivals → Sales → Inventory updates → Alerts → Synchronization
Resilience under changing conditions (traffic peaks, weather, connectivity failures)	

Simulation Stress-Testing Scenarios

Three representative environments were simulated:

- **Baseline (Status Quo):** Standard mall operation with no adaptive mechanisms. Customer traffic follows natural weekly patterns, inventory managed reactively, and connectivity remains stable throughout. Used as the reference benchmark for performance comparison.

- **Active Optimization:** Activation of predictive restocking algorithms and dynamic promotional triggers responding to traffic peaks and low-conversion periods, improving buyer conversion rates and operational throughput.

- **Failure Mode (Systemic Stress):** Environment characterized by intermittent connectivity disruptions, synchronization delays between nodes, and degraded inventory alert responsiveness, simulating real-world infrastructure instability.

Comparative Scenario Analysis

Metric	Baseline	Optimization	Failure
Cumulative Sales	~ 510	~ 1200	~420
Sync Delay	Minimal	Zero	0 - 8 (frequent)
Inventory Stability	High	Dynamic	Low / Erratic
Buyer Conversion	Low	Best	Near Zero
Response Time	1.0 - 1.9	0.6 - 1,5	Degraded

Saturdays always had the most visitors. Only the Optimization scenario successfully converted that traffic into sales.

Design Validation and Complexity Insights

The simulation results systematically confirm the soundness of design decisions established across the previous three workshops:

Synchronization Architecture Validated:

The failure scenario demonstrated conclusively that connectivity interruptions cascade into degraded performance across all system components simultaneously - confirming that offline synchronization with local data caching must be treated as a first-priority implementation requirement rather than an optional enhancement.

Modular Architecture Confirmed

The modular system structure proposed in Workshop 2 proved operationally sound, allowing independent components to function in an integrated manner while preserving the flexibility, scalability, and extensibility required for phased deployment and future expansion.

Phased Implementation Strategy Supported:

Results from the optimization scenario confirm that the complexity and interdependency of adaptive mechanisms justify the phased rollout strategy defined in Workshop 3, deploying core infrastructure before progressively introducing predictive and promotional capabilities.

Emergent Behaviors Identified:

The combination of predictive restocking and dynamic promotions generated sales results far exceeding either mechanism in isolation, highlighting the importance of analyzing the system as an integrated whole rather than as a collection of independent functional modules.

Recommendations

What to Prioritize Before Deployment

1. Strengthen Synchronization

Offline-first architecture with local caching. Connectivity is the system's single biggest vulnerability.

2. Adjust Replenishment Policies

Use predictive mechanisms to anticipate high-demand periods. Avoid stockouts especially on weekend peaks.

3. Deploy Promotions Early

Simulation showed promotions directly increase conversion. Don't save this for Phase 3 - introduce in Phase 2.

4. Saturday Campaigns

Design specific promotional triggers for Saturdays. Highest foot traffic day needs targeted conversion strategy.

Future Work
What Comes Next

- Loyalty programs & customer facing features
- Cross-mall benchmarking
- Improved ML forecasting as real data accumulates
- Extended simulation with longer time horizons and more store types

Conclusions

From Problem to Validated Solution

- **What the simulation proved:** Optimization features more than doubled sales vs baseline. The system works - when connectivity holds.
- **Biggest vulnerability:** Connectivity; When sync fails, inventory control, sales processing, and response times all degrade together. Offline sync must be solved before deployment.
- **The persistent gap:** High foot traffic ≠ high sales. The platform's engagement tools (promotions, recommendations) are not optional features - they are core to the value proposition.

Thank You

Better design, better living.